

CHIPPEWA COUNTY AP, MI, USA

WMO: 727344

Lat: 46.250N

Lon: 84.467W

Elev: 799

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04869

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -5.3	(d) -20.1	(e) 1.9	(f) -7.0	(g) -14.7	(h) 2.6	(i) -3.4	(j) 26.4	(k) 22.2	(l) 24.1	(m) 23.1	(n) 6.0	(o) 80	(p) 0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	83.0	69.2	81.0	67.6	78.6	65.6	71.9	80.1	69.7	76.8	67.9	75.0	9.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.7	112.7	76.9	66.6	100.9	73.5	65.4	96.5	72.1	36.1	80.5	34.3	77.6	32.6	74.6	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	20.6	18.6	-16.1	88.8	6.6	4.1	-20.8	91.8	-24.7	94.2	-28.4	96.5	-33.2	99.5
			-16.5	74.9	5.9	1.7	-20.7	76.1	-24.2	77.0	-27.4	78.0	-31.7	79.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	16.1	17.3	25.9	38.4	51.6	61.1	65.7	64.6	57.7	45.5	33.5	22.5
	DBStd	19.79	11.55	10.63	10.70	8.57	8.32	6.98	5.86	5.33	7.58	8.17	8.97	10.21
	HDD50	4719	1050	915	751	361	80	5	0	0	17	193	496	851
	HDD65	8683	1515	1335	1214	799	422	153	64	73	241	607	944	1316
	CDD50	1723	0	0	3	11	130	338	486	452	248	52	3	0
	CDD65	212	0	0	0	0	7	36	85	61	22	1	0	0
	CDH74	1618	0	0	3	3	102	350	661	369	125	5	0	0
	CDH80	304	0	0	0	0	21	72	141	50	20	0	0	0
(14) Wind	WSAvg	8.5	8.5	9.1	9.2	10.1	9.5	8.1	7.4	7.3	7.6	8.4	8.7	8.5
(15) Precipitation	PrecAvg	35.0	2.3	1.6	2.0	2.5	2.8	3.1	3.1	3.1	4.2	3.9	3.5	2.9
	PrecMax	47.6	4.3	2.9	4.7	5.2	5.1	6.0	7.0	4.9	8.7	7.2	6.8	5.9
	PrecMin	25.1	0.9	0.3	0.2	1.1	0.8	0.8	1.2	1.5	0.7	1.2	1.3	0.3
	PrecStd	5.4	0.8	0.7	1.1	1.1	1.1	1.1	1.4	1.0	1.8	1.6	1.3	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	45.1	60.7	70.3	82.2	86.3	88.1	84.2	82.4	73.0	60.8	48.1
		MCWB	39.7	40.6	48.9	55.1	65.2	70.0	70.9	70.6	69.4	62.5	53.7	46.5
	2%	DB	36.7	37.6	50.3	63.3	75.6	81.7	83.7	81.3	77.6	66.4	53.8	42.1
		MCWB	35.3	35.1	44.1	49.5	60.3	67.7	70.1	68.7	66.1	59.2	49.4	39.3
	5%	DB	34.5	35.6	45.1	57.0	72.2	78.7	81.1	78.8	73.4	62.8	48.4	38.2
		MCWB	33.2	33.5	38.9	44.9	57.7	64.6	67.8	66.6	64.3	56.6	45.5	36.3
	10%	DB	32.1	32.3	39.3	52.2	66.4	74.5	78.0	75.0	70.4	57.3	45.7	36.2
		MCWB	31.1	30.7	35.0	43.0	55.1	62.3	65.7	64.8	62.9	52.2	42.9	34.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.7	41.7	52.5	57.1	67.8	72.9	74.9	73.3	72.0	65.5	54.4	46.2
		MCDB	42.0	44.8	56.1	67.4	78.4	82.5	83.1	81.2	78.5	69.1	58.3	46.7
	2%	WB	35.6	36.0	45.2	51.0	63.9	69.4	72.2	71.0	68.9	60.6	49.5	39.4
		MCDB	36.5	37.9	49.3	60.3	72.8	78.0	80.5	77.9	74.5	66.2	53.2	41.2
	5%	WB	33.4	32.9	39.3	46.6	60.0	66.9	69.9	68.8	66.5	56.9	46.1	36.8
		MCDB	34.5	34.7	43.8	54.5	67.9	75.2	77.5	75.3	71.9	61.7	48.1	38.3
	10%	WB	30.8	30.5	36.0	43.5	56.4	64.4	67.7	66.9	64.1	53.6	43.3	34.5
		MCDB	31.9	32.0	39.5	51.7	64.6	71.3	75.1	73.3	68.7	57.1	46.1	35.8
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	15.2	16.5	16.9	20.4	20.1	19.5	18.2	17.2	14.4	11.8	12.3
		MCDBR	11.7	14.7	20.0	26.5	27.7	25.1	22.9	21.1	20.1	19.9	16.2	13.3
		MCWBR	10.9	12.4	13.7	16.0	15.0	13.8	11.8	11.1	11.2	13.4	13.0	11.8
	5% WB	MCDBR	11.6	14.4	18.6	23.6	24.0	21.1	19.7	18.1	18.3	18.2	15.1	12.5
		MCWBR	11.0	12.7	13.7	15.8	15.1	12.9	11.8	11.0	11.6	13.5	12.9	11.5
(40) Clear-Sky Solar Irradiance	taub		0.284	0.290	0.316	0.353	0.367	0.386	0.386	0.376	0.361	0.329	0.300	0.277
	taud		2.353	2.330	2.363	2.336	2.376	2.350	2.368	2.395	2.432	2.510	2.516	2.427
	Ebn at Noon		259	281	289	286	285	279	277	276	271	266	254	250
	Edh at Noon		23	29	33	37	37	38	37	35	31	25	20	20
(44) All-Sky Solar Radiation	RadAvg		347	639	1055	1426	1771	1912	1923	1628	1221	711	373	266
	RadStd		37	68	117	158	146	132	99	105	94	82	49	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page